

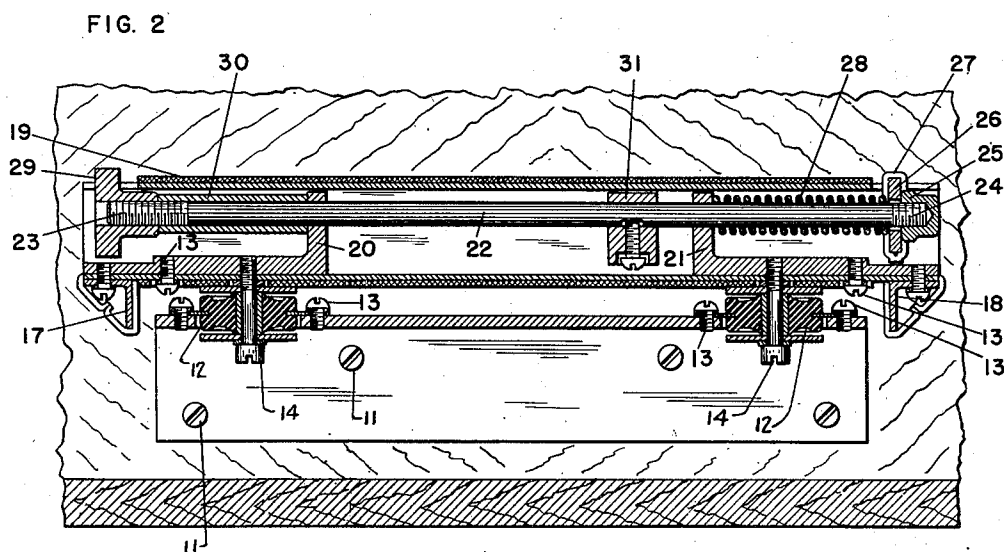
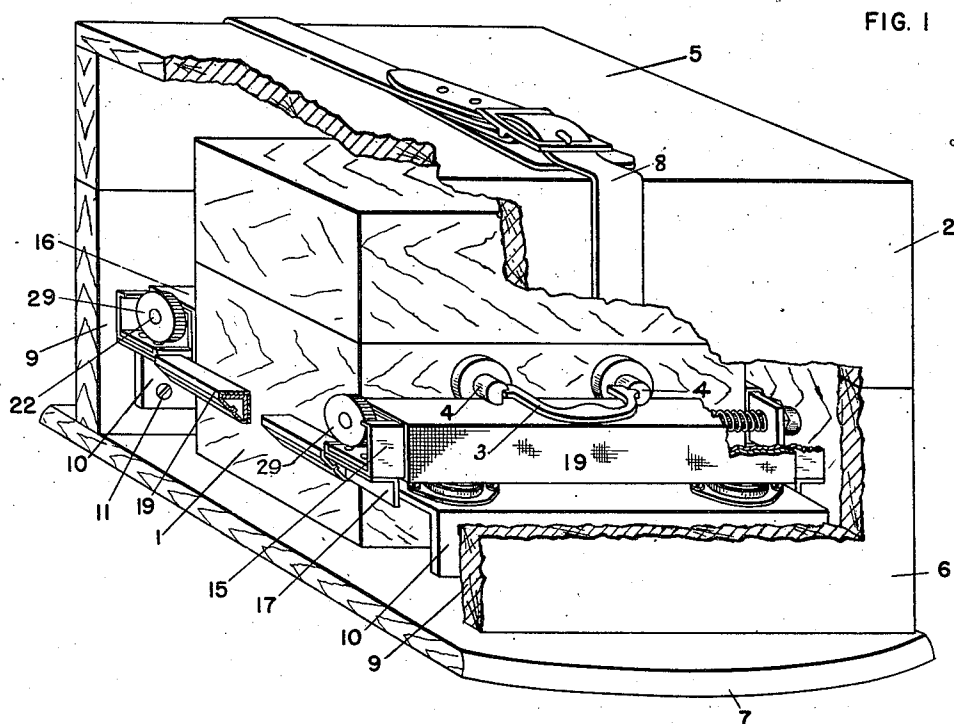
Dec. 30, 1947.

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2,433,509

CHRONOMETER BOX SECURING SUPPORT

Filed Feb. 21, 1945



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UNITED STATES PATENT OFFICE

2,433,509

CHRONOMETER BOX SECURING SUPPORT

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Application February 21, 1945, Serial No. 579,144

3 Claims. (Cl. 248—358)

1

This invention relates to an apparatus for mounting a chronometer box containing a chronometer in such a manner that it can be securely held in a resiliently mounted support at all times and easily and quickly released so that the chronometer and box may be easily and quickly removed.

The object of the present invention is to provide an apparatus for mounting a chronometer box supporting a chronometer in a carrying case in such a manner as to isolate said chronometer box from vibrations and protect it against shocks.

A further object of the present invention is to provide a support for a chronometer box which will securely hold the box in the desired position and from which support the box may be speedily and easily removed.

A further object of the present invention is to provide a support for a chronometer box which will rigidly grip said box without mutilating or penetrating the box in any way.

A further object of the present invention is to provide a support for a chronometer box which, when once mounted in a carrying case, will not have any detachable parts yet will permit the chronometer box to be easily secured and quickly removed.

A still further object of the present invention is to provide a chronometer box support which may be housed entirely within the carrying case, which provides a vibration and shock resisting mounting for the box and which will permit quick and easy removal of the chronometer box and maintain its indicator within the the carrying case.

A still further object of the present invention is to provide a support which substantially encircles a chronometer box, which rigidly grips said box in a resiliently supported suspension and from which the box may be quickly and easily removed.

With the foregoing and other objects in view, the invention resides in the novel arrangement and combination of parts and in the details of construction hereinafter described and claimed, it being understood that changes in the precise embodiment of the invention herein described may be made within the scope of what is claimed without departing from the spirit of the invention.

The invention is illustrated in the accompanying drawing in which:

Fig. 1 is a broken perspective view showing the invention with the chronometer box in position;

2

Fig. 2 is a cross section showing the details of the supporting member.

In the past, chronometer boxes have been carried in a padded carrying case generally lined with felt or some similar surface abrasion preventing material. This padded felt gave no protection against the vibration or shock which a chronometer in transport is bound to be subject to. In the present day transportation where exclusive vibration and shock caused by road conditions, sudden stops or plane landings, the protection of a chronometer from that vibration or shock is a necessity if the instrument is to function correctly. It is to provide protection against vibration and shock while in transport, as well as to prevent abrasion of the chronometer box that the present support is designed.

With marine chronometers the chronometer itself is mounted in gimbals which in turn are supported in a chronometer box which chronometer box is itself supported in a carrying case. This is clearly illustrated in Fig. 1 which, however, does not show the chronometer or the supporting gimbals, as they form no part of this invention but which does show the chronometer box 1 mounted in the carrying case 2.

The chronometer box 1 is customarily constructed with handles 3 located on opposite sides of the box and formed with a hand grip pivotally mounted in posts 4 which extend outward from the sides of the box. These supports allow the hand grip to be swung above the lower line of the pivotal posts 4.

The carrying case 2 is of the usual construction having a hinged top 5 and a lower section 6 mounted on a base 7 and provided with a carrying strap 8 and a lock, which is not shown as it forms no part of this invention. Rigidly secured to the sides 9 of the lower section 6 are angle brackets 10 upon which the chronometer box support member is mounted.

The angle brackets 10 are secured to the sides 9 by screws 11 and have mounted thereon vibration or shock absorbing members 12 which are secured to the angle brackets by screws 13. These vibration absorbing members are a well known product and may be extremely sensitive or subject to resist a heavy shock. They form no part of the invention except to provide a support which will resist vibration.

Secured to the rubber supports by bolts 14 are squared tubes 15 and 16 which are connected to a front angle brace and a rear angle brace 17 and 18 respectively to form a hollow square to receive the box 1. Each of the square tubes and the

3

front and rear braces is covered with cloth or felt 19 to prevent marring the finish of the box. Within the squared tubes 15 and 16 are a pair of supporting bearings 20 and 21 in which is slidably mounted a rod 22. These bearings are secured in position within the tubes 15 and 16 by the screws 13'.

The rods 22 are threaded at each end as at 23 and 24, the rear end 24 receiving a cap nut 25 which bears against a clamping bar 26 loosely carried by the rod 22. Clamping bar 26 has a felt covering 27 similar to the coverings of the tubes and the angle braces and is held securely against the cap nut 25 by a coil compression spring 28 extending between the clamping bar and the rear bearing 21. The threaded end 23 of the rod is designed to receive a knurled nut 29 which bears against a sleeve 30 and through the sleeve to the stationary forward bearing 20. It will be easily seen that adjustment of the knurled nut causes the clamping bar 26 to be moved back and forth with the rods 22. The lugs 31 are keyed to the rods 22 and being of a square formation prevent rotation of the rods 22 upon movement of the knurled nut 29.

In initially placing the box 1 within the carrying case 2 the box 1 is allowed to rest by the posts 4 contacting the tubes 15 and 16, the box being thus loosely held within the square formed by the tubes 15 and 16 and the front and rear angle braces 17 and 18. The knurled nut 29 is then turned on the rods 22 to move the clamping bar toward the front, said clamping bar securely engaging the rear of box 1 forcing the box into engagement with the felt covered front angle brace 17 and securely holding the box in the vibration absorbing supporting tubes. The reverse motion of the knurled nuts is all that is needed to almost instantly free the box from the clamping action of the rear bar making it possible to either speedily clamp the box 1 in supported position or to release it for withdrawal.

What is claimed is:

1. A vibration absorbing support for a box comprising a pair of opposed squared supporting tubes vibration absorbing mounts supporting said tubes, a front angle brace and a back angle brace extending between said tubes and permanently connected thereto, felt coverings for said tubes and said angle braces, a pair of supporting bearings carried in each of said tubes, a rod slidably

4

mounted in each of said pairs of bearings, a clamping bar carried by said rods and movable transversely of said tubes, knurled finger nuts threaded on said rods and bearing against the ends of said tubes to furnish movement to said rods, and a squared lug carried by said rod and within said tubes to prevent rotation of said tubes whereby tightening of said knurled nuts causes movement of said clamping bar to securely clamp said box between said front angle brace and said clamping bar.

2. A vibration absorbing support for a chronometer box comprising a pair of opposed hollow supporting members vibration absorbing mounts supporting said members, a pair of clamping members located at right angles to said supporting members, and means carried within said hollow members for moving one of said clamping members toward and away from the other clamping member to securely clamp a box carried on said supporting members.

3. A vibration absorbing support for a chronometer box comprising a pair of opposed hollow supporting members, vibration absorbing mounts supporting said members, a bar extending at right angles to and connecting said opposed supporting members, a clamping bar extending at right angles to and across said supporting members and means carried within said hollow supporting members for moving said last mentioned clamping bar toward and away from said first mentioned bar, to securely clamp a box carried on said supporting members.

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